

A surreal illustration with a woman's face in the center, large hands holding strings, a woman on a crescent moon, and a sleeping woman in a bed with a fox.

DREAMING OR AWAKE?

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Overview

- ◆ Goal: distinguish between dreaming and awake states of the brain.



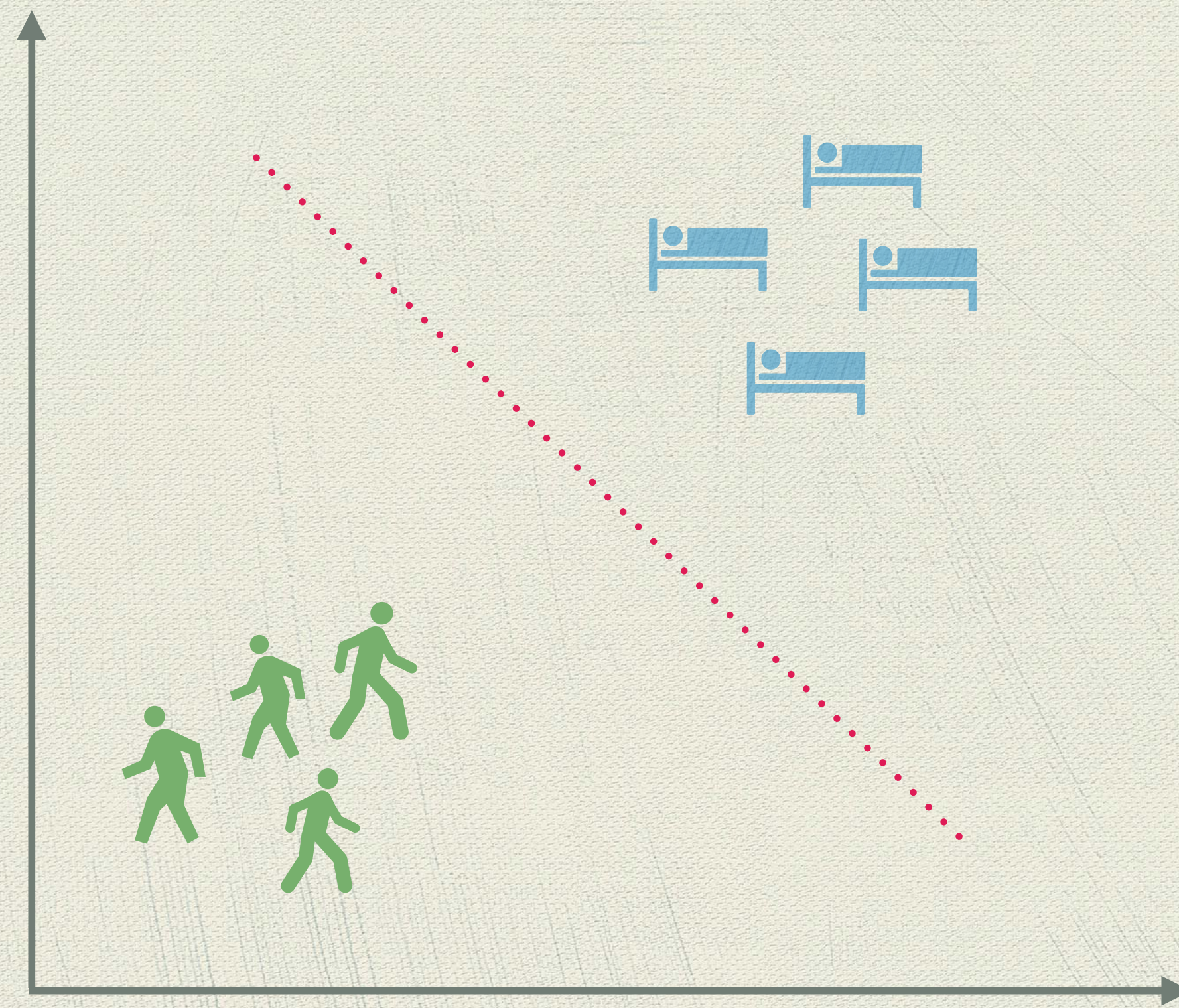
Motivation

- ◆ To better understand human brain (broad); to better understand recovery and loss of consciousness (ROC and LOC).
- ◆ Examples: surgery; anesthesia; sleep quality

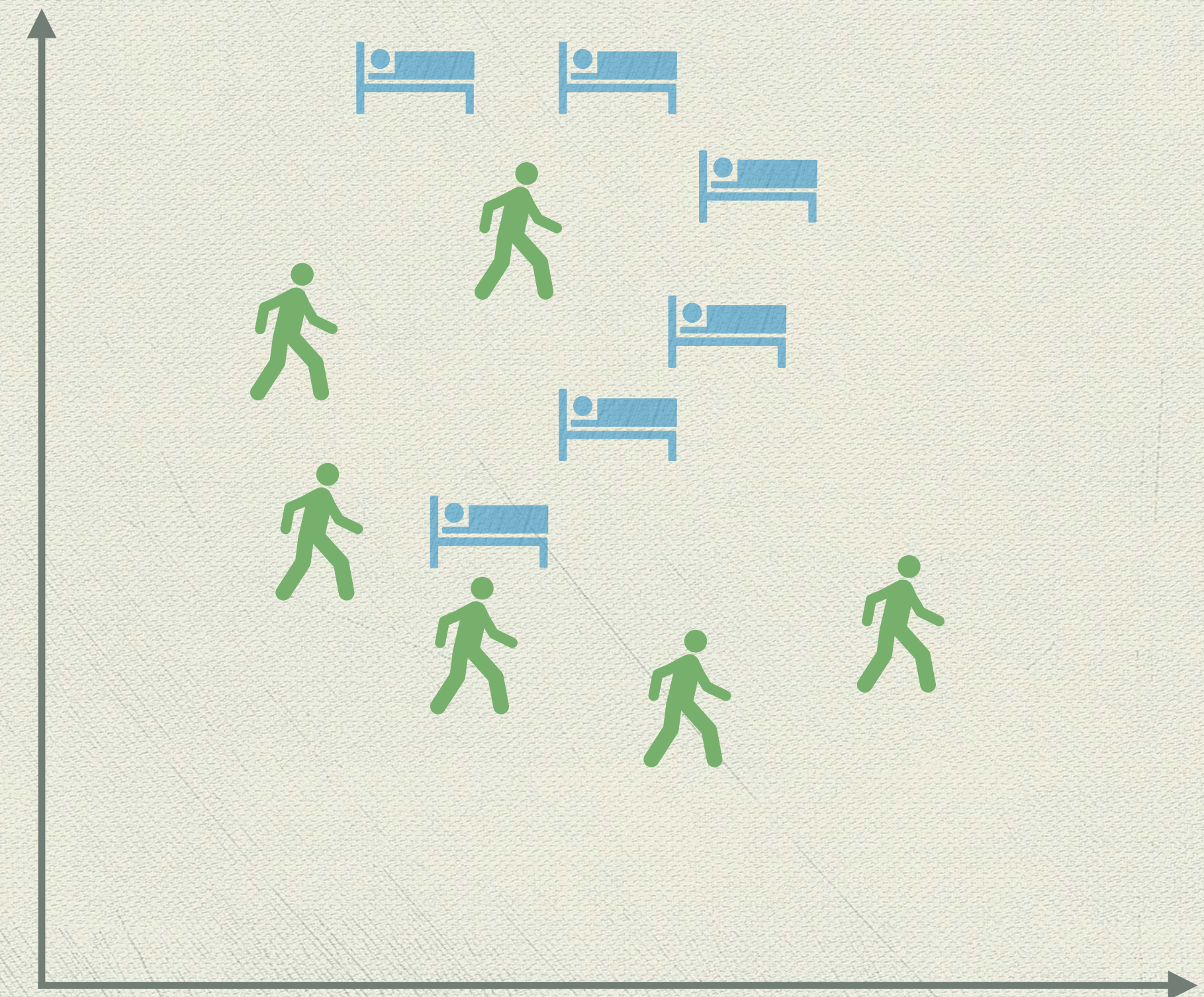


Motivation

- ◆ Standard algorithms don't work well on the current dataset.



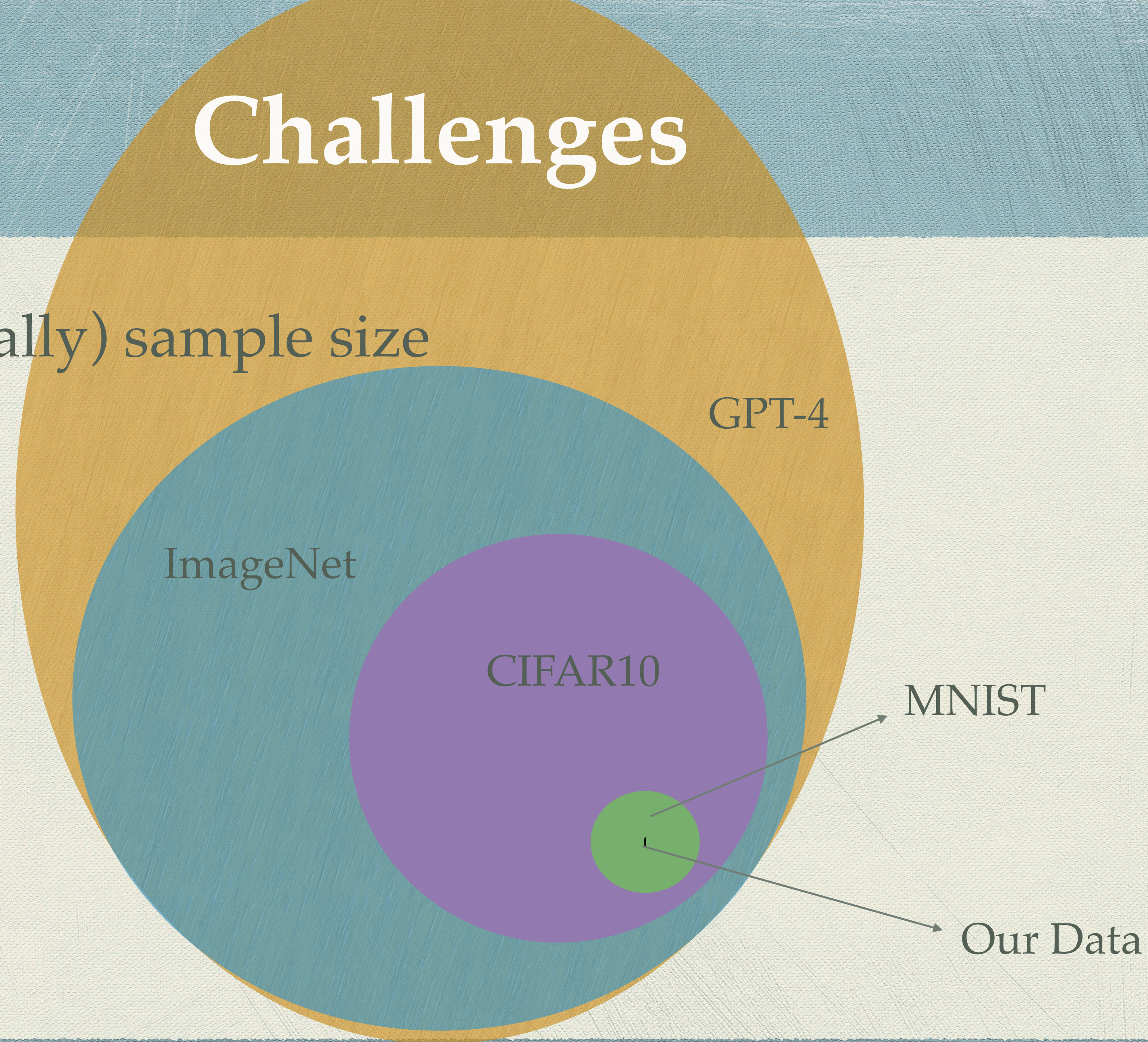
Expectation: clusters



Reality: swirled data

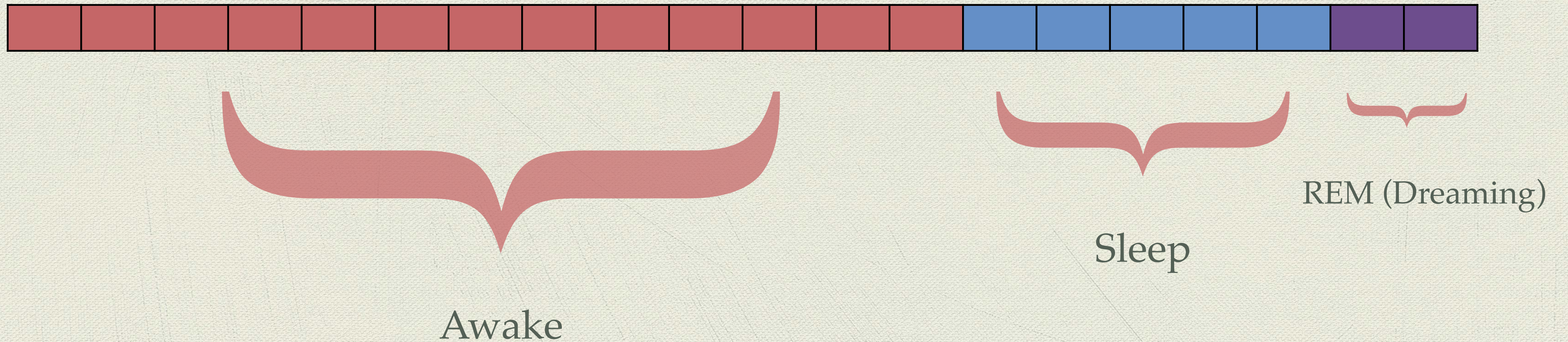
Challenges

- ◆ ...Small (literally) sample size
- ◆ (n=33)



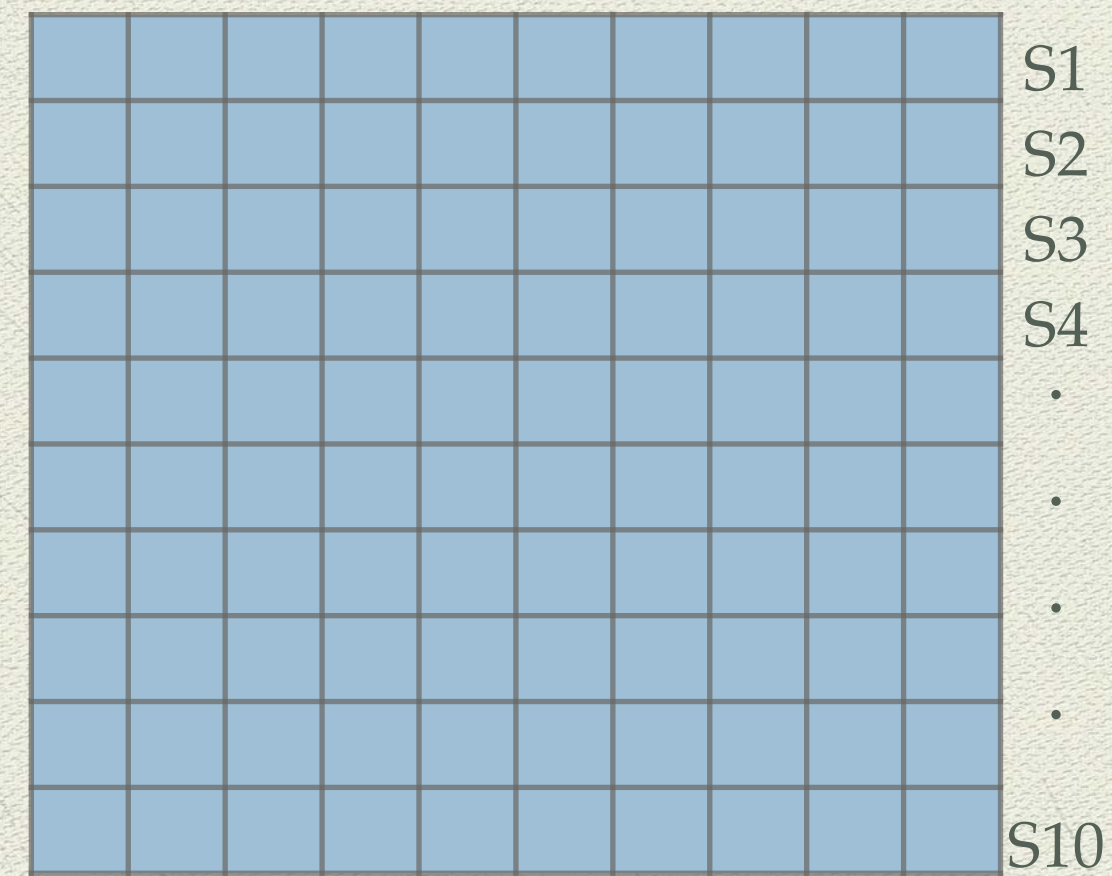
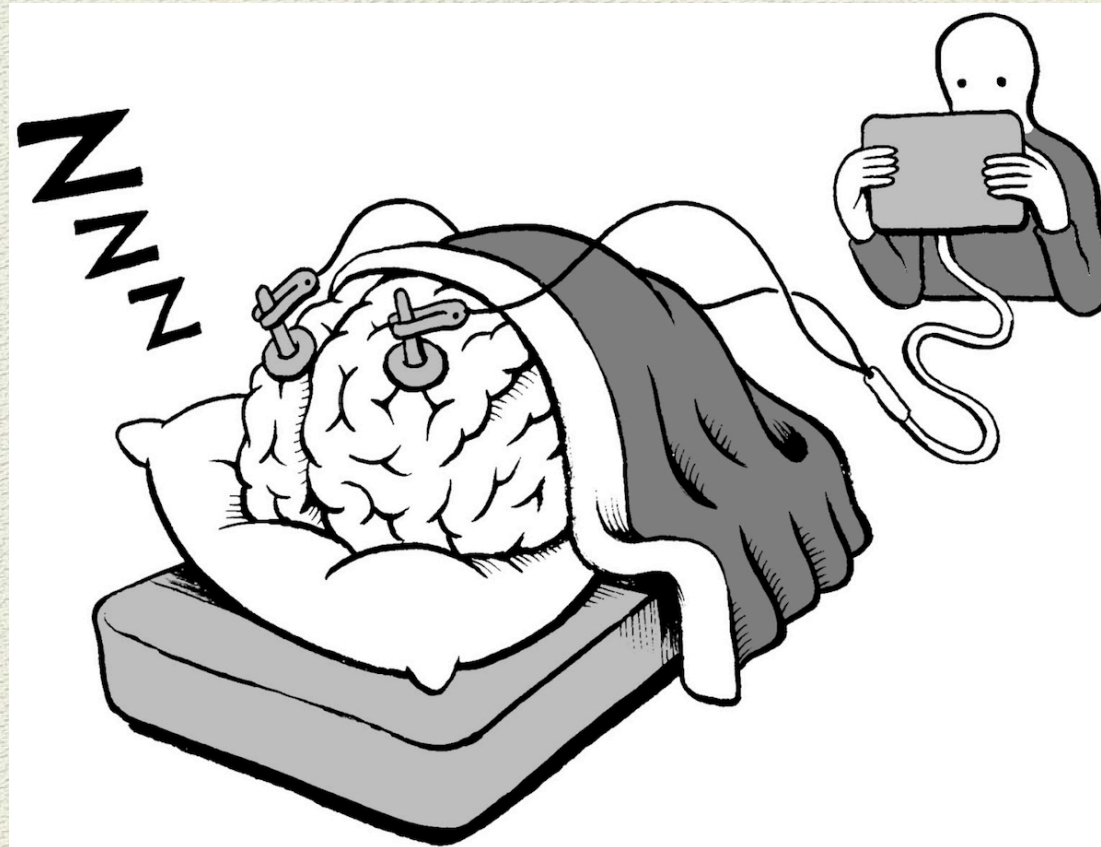
Challenges

◆ Unbalanced labels

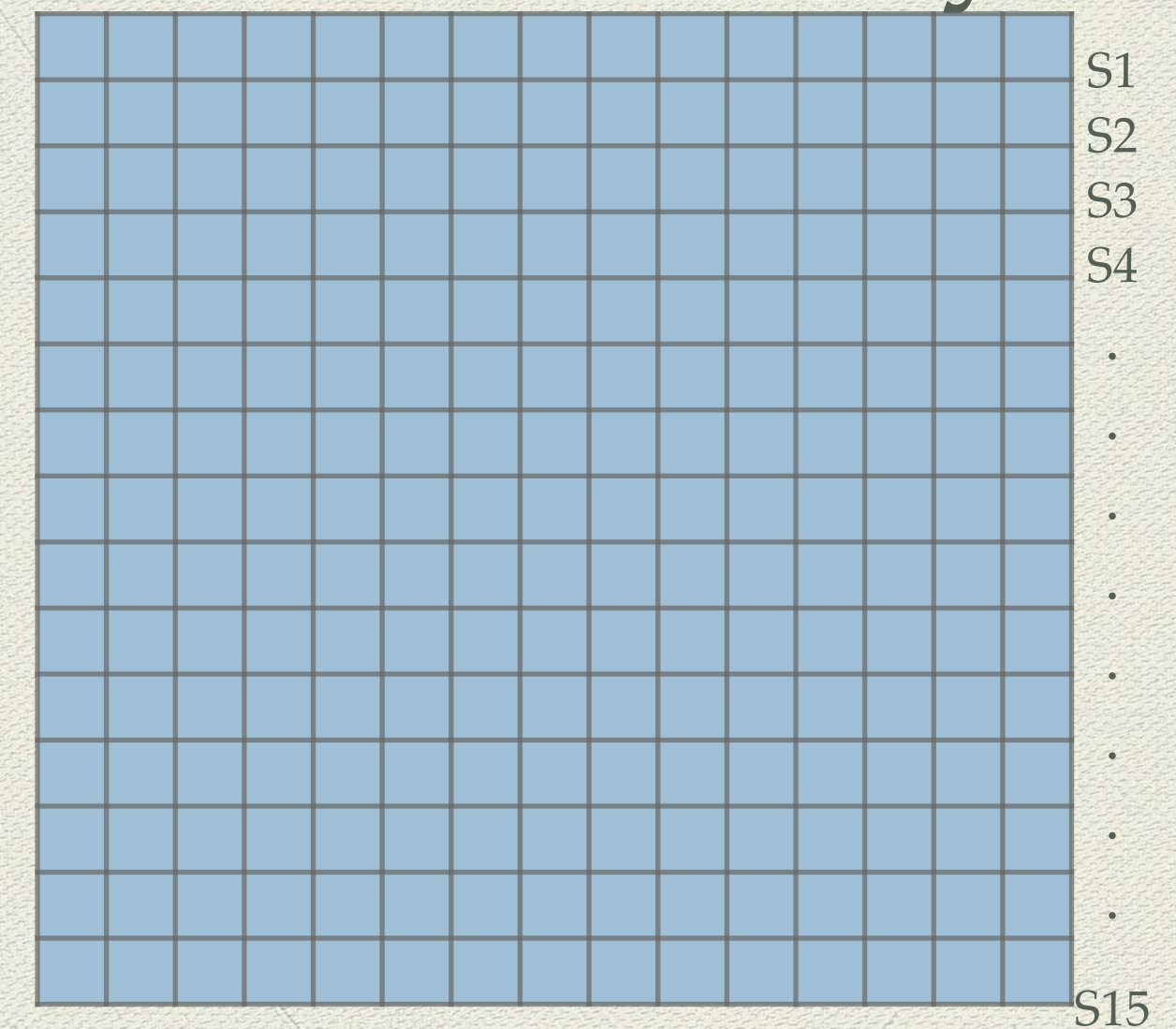


Challenges

- ◆ Electroencephalogram (EEG): recording of brain activity



Subject A with signals
from 10 sensors



Subject B with signals
from 15 sensors

- ◆ Correlation Map

Challenges

- Missing signals in certain brain regions

Prefrontal Lobe

Prefrontal Lobe	1	?	?	?	?	?	?	?	?
	?								
	?								
	?								
	?								
	?								
	?								
	?								
	?								
	?								

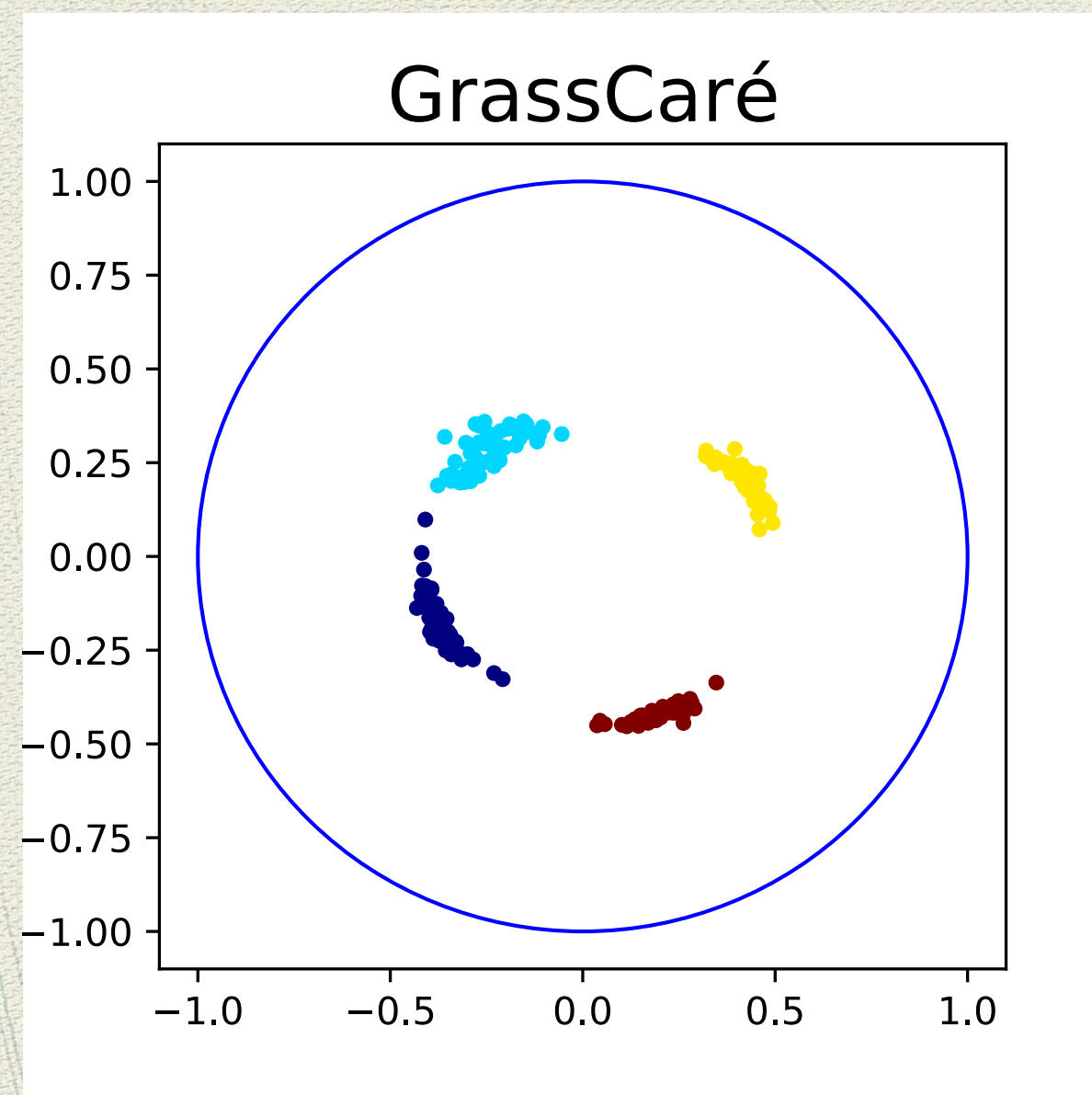
		Parietal Lobe				Temporal Lobe			
			?				?		
Parietal Lobe	?	1	?	?	?	?	?	?	?
		?					?		
		?					?		
		?					?		
		?					?		
Temporal Lobe	?	?	?	?	?	?	1	?	?
		?					?		
		?					?		
		?					?		

Our Idea

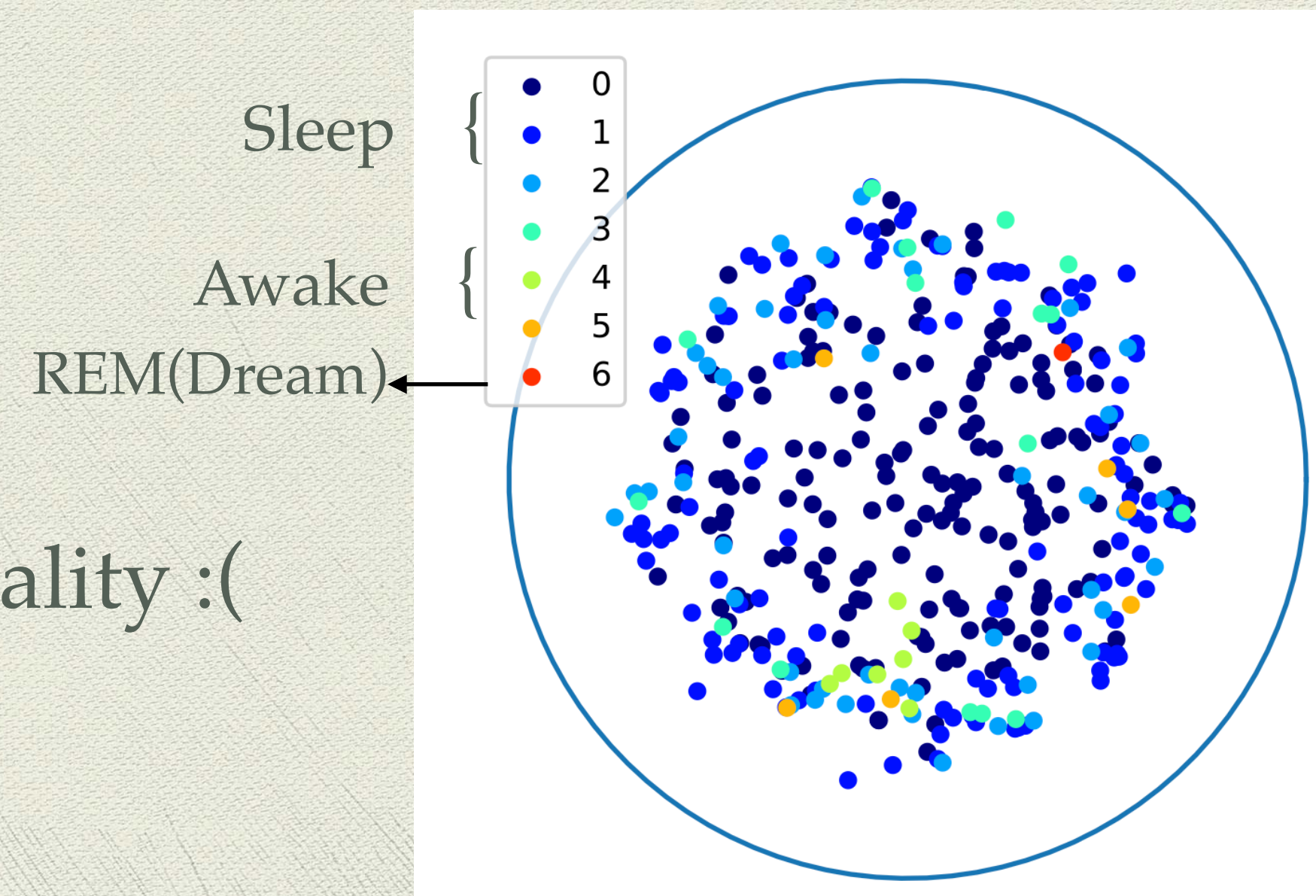
- ◆ Preliminary Analysis

- ◆ Robust PCA + GrassCare (A subspace visualization technique)

Ideal Case :)



Reality :(



Acknowledgment

- ◆ Thanks for data from Professor Matthew Bank's lab, Department of Neuroscience, UW-Madison
- ◆ Thanks for the match by Research Bazaar (Michael Ferris et al.)
- ◆ Thanks for everyone's attention!